



Calcium isotope constraints on a Middle Ordovician carbon isotope excursion

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ARTICLE INFO

Edited by: Prof A Jacobson

Keywords:

Calcium isotopes
Carbon isotopes
Carbonate diagenesis
Ordovician

ABSTRACT

The Middle Ordovician Darriwilian Stage (~469 – 458 Ma) records a ~2‰ positive carbon isotope shift known as the MDICE (Mid-Darriwilian Carbon Isotope Excursion). Although studies have shown that the MDICE is a globally synchronous event, the link between the MDICE and changes in the global carbon cycle remains unclear. This is largely due to possible local processes including diagenesis and variations in carbonate polymorphs that can obscure $\delta^{13}\text{C}$ signals recorded in shallow marine carbonate rock from the global dissolved inorganic carbon reservoir. Here, we use paired measurements of $\delta^{13}\text{C}$, $\delta^{44/40}\text{Ca}$, and Sr/Ca from a stratigraphic section at Meiklejohn Peak in southwest Nevada (USA) to constrain the potential for local processes in decoupling the recorded $\delta^{13}\text{C}$ signals from the global carbon cycle. We find that variations in $\delta^{44/40}\text{Ca}$ and Sr/Ca in this section at Meiklejohn Peak can in part be explained by changes in primary mineralogy and diagenesis. However, these processes are unable to reproduce the entire shift observed in $\delta^{13}\text{C}$, with the remaining portion of the $\delta^{13}\text{C}$ curve driven by either local changes in platform water mass dissolved inorganic carbon (DIC) or global carbon cycling. While we cannot completely deconvolve local signals from the recorded $\delta^{13}\text{C}$, we argue that the MDICE may, at least in part, reflect a primary signal of increase organic carbon burial related to changes in primary productivity and nutrient delivery during increased basaltic weathering associated with the Taconic uplift. The proposed link between tectonic uplift and increase in primary productivity strengthens the notion that tectonic processes played a significant role in modulating changes in the global carbon cycle during the Ordovician Period.

1. Introduction

The carbon isotopic composition ($\delta^{13}\text{C}$) of ancient carbonate rocks has long been used as a proxy for changes in the global carbon cycle throughout Earth history (e.g., Berner, 2006; Katz et al., 2005; Knoll et al., 1986; Saltzman and Thomas, 2012). Kinetic isotopic fractionation during photosynthesis preferentially incorporates ^{12}C into organic matter, which leads to ^{13}C enrichment in the residual dissolved inorganic carbon (DIC) reservoir. Consequently, positive shifts in $\delta^{13}\text{C}$ values on time scales of millions of years (i.e., carbon isotope excursions or CIEs) in marine carbonate rocks are commonly interpreted as signals of an increase in the fraction of carbon buried as organic matter (f_{org}), which have implications for atmospheric carbon dioxide removal or oxygen production (e.g., Edwards et al., 2017; Kump and Arthur, 1999; Lenton et al., 2018).

A growing number of studies, however, have challenged this

canonical interpretation on the basis that $\delta^{13}\text{C}$ values recorded in shallow marine carbonates ($\delta^{13}\text{C}_{\text{carb}}$) are sensitive to local variations such as platform-scale local processes (e.g., diurnal carbon cycle, vital effects, and platform seawater chemistry gradient (Geyman and Maloof, 2019; Gischler et al., 2009; Patterson and Walter, 1994)), variations in primary mineralogy/carbonate polymorphs (Kimmig and Holmden, 2017), and post-depositional diagenesis (Ahm et al., 2018; Allan and Matthews, 1982; Higgins et al., 2018). First, variations in $\delta^{13}\text{C}_{\text{carb}}$ due to local processes produce a wide range of $\delta^{13}\text{C}$ values across different carbonate rock components and facies in modern environments (Geyman and Maloof, 2021). As such, variations in depositional environment in stratigraphic records can potentially manifest into chemostratigraphic changes in $\delta^{13}\text{C}_{\text{carb}}$ (Geyman and Maloof, 2021). Secondly, variations in primary mineralogy can affect the recorded $\delta^{13}\text{C}$ in shallow marine carbonate because aragonite minerals fractionate $\delta^{13}\text{C}$ ($\Delta^{13}\text{C} \sim 2.7\text{‰}$) more strongly than calcite ($\Delta^{13}\text{C} \sim 1.0\text{‰}$) during carbonate

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precipitation (Kimmig and Holmden, 2017; Romanek et al., 1992). Swart (2008), for instance, suggests that the gradual positive shift in $\delta^{13}\text{C}$ recorded in Neogene carbonate rocks from the peri-platform Bahamas is a result of increasing aragonite production and export from the platform top coupled with platform margin progradation during sea level fall and/or a shift in the dominant carbonate polymorph precipitation from low magnesium calcite to aragonite. Lastly, diagenesis can alter and reset the recorded $\delta^{13}\text{C}$ in carbonate sediments because diagenetic fluids often have distinct $\delta^{13}\text{C}$ values that are different from the platform DIC which the carbonate precipitated from, either as meteoric (Allan and Matthews, 1982; Melim et al., 2002) or open marine water (Higgins et al., 2018). Consequently, diagenetically altered carbonate rocks may preserve the $\delta^{13}\text{C}$ signature of the diagenetic fluid rather than the corresponding DIC (Ahm et al., 2021; Allan and Matthews, 1982; Crockford et al., 2021). It is noteworthy that, while many factors can play significant roles in decoupling the $\delta^{13}\text{C}$ of modern shallow water carbonate from global DIC, it has been hypothesized that heterogeneities due to platform-scale local processes are less well-expressed in ancient carbonate rocks due to apparent isotopic homogenization of bulk sediment (Geyman and Maloof, 2021; Husson et al., 2016). This homogenization process, which may be due to physical mixing (i.e., abrasion and intra-platform transport) and/or chemical exchange, may explain the fact that the differences in $\delta^{13}\text{C}$ between carbonate matrix and well-preserved fossils observed in Paleozoic rocks are less than the documented $\delta^{13}\text{C}$ variabilities among carbonate grains from the Bahamas (see Geyman and Maloof, 2021 for a more thorough discussion).

One potential approach to constrain the relative role of variations in mineralogy and post-depositional diagenesis in affecting $\delta^{13}\text{C}$ in carbonate rocks is using paired measurement of shallow water carbonate $\delta^{13}\text{C}$ with calcium isotope ($\delta^{44/40}\text{Ca}$) and Sr/Ca ratios. These geochemical measurements can provide a framework to assess the extent of diagenetic alteration in carbonate rocks (Ahm et al., 2018; Higgins et al., 2018; Husson et al., 2015; Lau et al., 2017; Lau and Hardisty, 2022) and variations in primary mineralogy (Gussone et al., 2005; Kimmig and Holmden, 2017). Slow processes such as recrystallization during carbonate diagenesis will reduce the fractionation factor between fluid and mineral which then shifts $\delta^{44/40}\text{Ca}$ of the carbonate rock toward fluid- or seawater-like values (i.e., higher $\delta^{44/40}\text{Ca}$). Similarly, due to rate effects on the Sr distribution coefficient, slow re-precipitation rates during diagenesis will drive down Sr concentrations (Fantle et al., 2020; Lau et al., 2017; Tang et al., 2008b). Additionally, aragonite fractionates $\delta^{44/40}\text{Ca}$ more strongly than calcite ($\Delta^{44/40}\text{Ca} \sim -1.5$ vs -0.9‰ ; Gussone et al. (2005)) and contains higher Sr content (Kinsman, 1969). Therefore, covariation between $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$ (and potentially Sr/Ca) can provide clues on the relative importance of early marine diagenesis and variations in primary carbonate mineralogy in driving shifts in $\delta^{13}\text{C}$. Besides, covariation between $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$ is unlikely to be driven by changes in seawater chemistry due to the longer residence time of Ca in the oceans compared to C (Griffith and Fantle, 2020). In addition, the impact of carbonate precipitation rate and kinetic effects can also produce covariation between these geochemical proxies (e.g., Wang et al., 2021, 2023).

Here we present paired measurements of $\delta^{44/40}\text{Ca}$ and Sr/Ca with $\delta^{13}\text{C}_{\text{carb}}$ from Middle Ordovician carbonate rocks from the Meiklejohn Peak section in southwest Nevada (USA) that preserve a positive shift in $\delta^{13}\text{C}_{\text{carb}}$, which has been proposed to be the Mid-Darriwilian Carbon Isotope Excursion (MDICE, Edwards and Saltzman, 2016). We find significant covariation between Sr/Ca – $\delta^{44/40}\text{Ca}$ and Sr/Ca – $\delta^{13}\text{C}$, but not between $\delta^{44/40}\text{Ca}$ – $\delta^{13}\text{C}$. These results and simple numerical modeling suggest that while variations in primary mineralogy and early marine diagenesis can partly explain the observed stratigraphic variations in Sr/Ca and $\delta^{44/40}\text{Ca}$ at Meiklejohn Peak, changes in the $\delta^{13}\text{C}$ of DIC (global ocean and/or platform water mass) are still required to fully explain the observed shift in $\delta^{13}\text{C}_{\text{carb}}$.

2. Background

2.1. The Mid-Darriwilian carbon isotope excursion

The Mid-Darriwilian carbon isotope excursion (MDICE) is a positive shift in $\delta^{13}\text{C}$ that has been documented in carbonate rocks (e.g., Ainsaar et al., 2010; Albanesi et al., 2013; Saltzman and Edwards, 2017; Wu et al., 2017) and well-preserved brachiopod shells (Rasmussen et al., 2016) in numerous localities across the globe. The magnitude of this excursion ranges from 0.5‰ (e.g., the Las Chacritas Formation in Argentina) to 2.5‰ (Pinesburg Station and Row Park formations in Maryland, USA), with an average value around 1.5‰ (see Fig. S1 for comparison). Conodont biostratigraphy (e.g., Albanesi et al., 2013; Wu et al., 2017) and strontium isotope chemostratigraphy (Saltzman and Edwards, 2017) suggest that the MDICE is a globally synchronous event (within the resolution of biostratigraphy and strontium isotope stratigraphy; ± 1 Myr). The chemostratigraphic expression of MDICE is marked by a gradual $\delta^{13}\text{C}$ increase ($\sim 0.75\text{‰/Myr}$) that started within the *Eoplacognathus pseudoplanus* (in Baltica) or *Histiodela holodentata* (in North America) biozones, reached its peak within the *Eoplacognathus suecicus* Zone, and gradually decreased toward pre-excursion baseline at the *Pygodus serra* Zone (e.g., Albanesi et al., 2013; Wu et al., 2017). Numerical age calibration to Geologic Time Scale 2020 (Cramer and Jarvis, 2020) suggests that this event lasted for ~ 5 million years. Additionally, while studies have speculated on links among the MDICE, carbon cycling, and global climate (e.g., Rasmussen et al., 2016), there is no consensus on what might have caused this event.

2.2. Geologic background and stratigraphy

Carbonate rock samples were collected from Meiklejohn Peak in southwest Nevada, which was located in the northwest margin of the Laurentian paleo-continent during the Ordovician Period (Fig. 1). The Ordovician rocks of western Laurentia were deposited in a shallow marine passive margin setting (Ross et al., 1989). Paleogeographic reconstructions place this region in the tropics during the Ordovician (Swanson-Hysell and Macdonald, 2017). The lithostratigraphic unit sampled for this study is the Antelope Valley Limestone.

The Antelope Valley Limestone is divided into three sub-units: the Lower, Middle, and Upper members. The Lower Member consists of mudstone, wackestone, and some packstone. This unit is interpreted to have been deposited in a subtidal environment between shallow oncolite shoals and deep slope deposits (Ross et al., 1989). The contact between the Lower Member and Middle Member is placed at the base of the occurrence of massively bedded nodular packstone of the Middle Member (Diamond, 2021). The Middle Member is composed of massive oncologic packstone and grainstone with sporadic traces of burrows (Fig. 1). This unit is interpreted as a shoal deposit which marks the paleo-shelf edge position of the Ordovician carbonate platform (Ross et al., 1989). The contact between the Middle and Upper Members is placed at the last occurrence of a massive packstone, which underlies a silty gray lime mudstone of the Upper Member (Diamond, 2021). The Upper Member is characterized by thinner bedded homogeneous dark grey mudstone. Some intervals of this unit contain thin silty micrite beds cyclically interbedded with cleaner micrites forming “ribbon” limestone. The middle portion of the Upper Member contains cyclic thin-bedded burrowed packstone and massive packstone (Fig. 1) which underlies sandy fossiliferous packstone (Fig. 1). This unit is overlain by the Eureka Quartzite. Ross et al. (1989) suggest that the Upper Member was deposited in a subtidal to nearshore environment.

The chronostratigraphic framework of the Meiklejohn Peak section is largely based on conodont biostratigraphy and $\delta^{13}\text{C}$ chemostratigraphy. Previous work by Harris et al. (1979) has established a biozonation division of this section, based on the appearance of North American and North Atlantic index fossils such as: *Histiodela holodentata*, *Eoplacognathus suecicus*, *Phragmodus polonicus*, *Polyplacognathus sweeti*

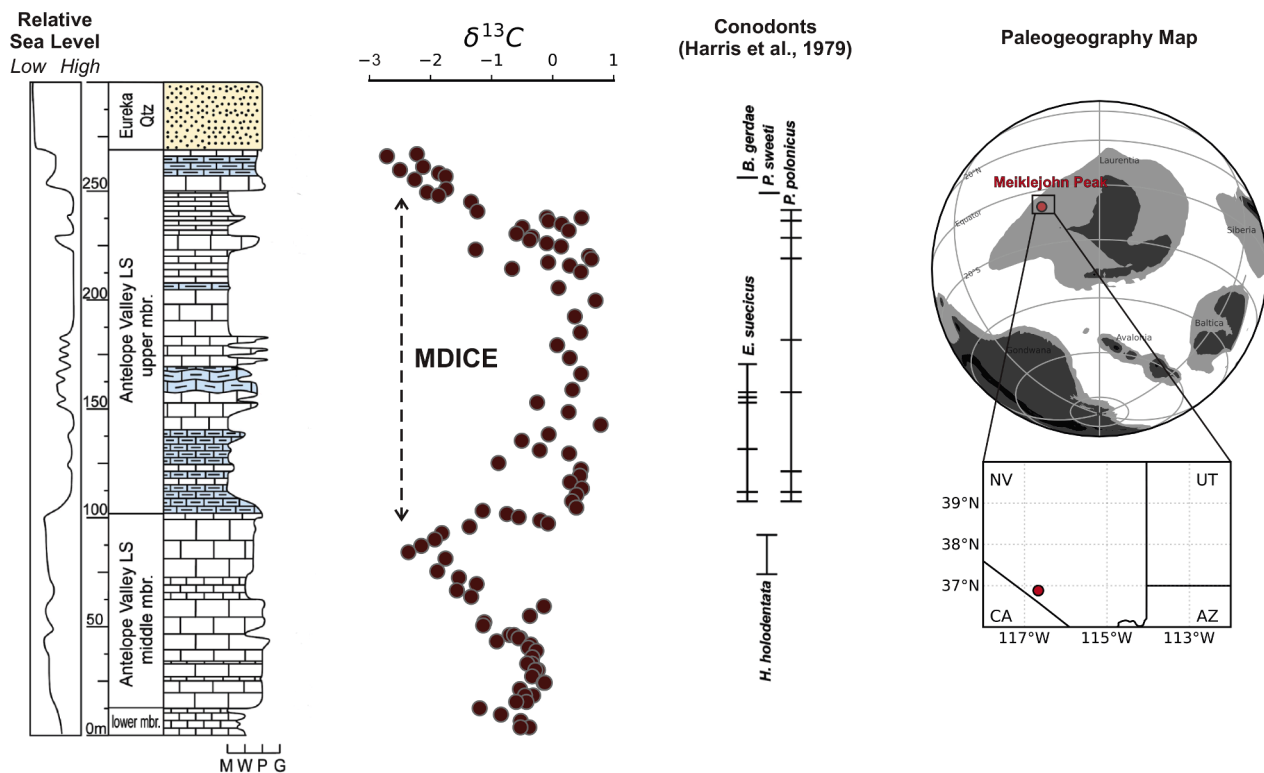


Fig. 1. Stratigraphy and paleogeography map of the Meiklejohn Peak section. MDICE = Mid-Darriwilian Carbon Isotope Excursion. Lithostratigraphic and $\delta^{13}\text{C}$ (‰ V-PDB) data are from Diamond (2021). LS = Limestone. Qtz = Quartzite. M, W, P, and G refer to mudstone, wackestone, packstone, and grainstone respectively. Conodont range is from Harris et al. (1979) (non-abbreviated species names *Histiodela holodontata*, *Eoplacognathus suecicus*, *Phragmodius polonicus*, *Polyplocognathus sweeti*, and *Baltoniodus gerdae*). Paleogeography map (Scotese, and Wright, 2018) shows that the Meiklejohn Peak section was on the northwest margin of the Laurentia continent during the Middle Ordovician. Grey shades shown in the paleogeography map reflect the reconstructed water depth and elevation. Light grey corresponds to water depth between 0 to 500 meters. Abbreviated state names: CA: California; NV: Nevada, UT: Utah; AZ: Arizona.

(*Cahabagnathus sweeti*), and *Baltoniodus gerdae* (Fig. 1). This zonation assigns most of the units exposed at Meiklejohn Peak (Lower, Middle, and most of Upper Member) as part of the Darriwilian Stage of the Middle Ordovician Series and the very top of the section as part of the Sandbian Stage of the Upper Ordovician Series. Specifically, the presence of *E. suecicus* zone in this section, which coincides with the most positive values of $\delta^{13}\text{C}_{\text{carb}}$, is consistent with the peak of the MDICE reported from stratigraphic sections in the Baltic (Wu et al., 2017). Notably, the lack of corroborating biostratigraphic constraints at the rising and declining limbs of this positive shift makes the chronostratigraphic context of this event less certain (Edwards and Saltzman, 2016). Previous studies of $^{87}\text{Sr}/^{86}\text{Sr}$ chemostratigraphy to constrain the age of these strata, using both conodonts and bulk carbonate, are consistent with the $\delta^{13}\text{C}_{\text{carb}}$ shift being correlated with the MDICE (e.g., Edwards et al., 2015; Saltzman et al., 2014). Numerical age calibration to Geologic Time Scale 2020 (Goldman et al., 2020) suggests that the Meiklejohn Peak section sampled in this study represents 11 Myrs time (see Fig. S1) and an average sedimentation rate of ~ 33 meter/Myr (higher sedimentation rates than sections in Baltica 5 – 7 meter/Myr).

3. Method

3.1. Sample preparation and column chromatography

All measurements done in this study were performed on carbonate rocks that previously had been measured for $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, and $^{87}\text{Sr}/^{86}\text{Sr}$ (Edwards et al., 2015; Diamond, 2021). Samples were prepared by drilling a fresh surface of polished carbonate rocks by hand while avoiding obvious veins, siliciclastic lithoclasts, and weathered surfaces. Approximately 10 mg of powdered carbonate rocks were weighed,

digested with 10 mL of 10 % ultrapure acetic acid for 24 h, and centrifuged to separate insoluble materials from solutions. A portion of the solutions (100 μL) was removed and diluted for major and trace element concentration measurements (Ca, Sr, Mg, Mn, Al) on a Perkin-Elmer Optima 4300DV Inductively Coupled Plasma Optical Emission Spectrometer in the Trace Elements Research Laboratory at Ohio State University. Elemental concentrations were determined using a series of in-house standard solutions, made by serial dilution of purchased standards. Major and trace element abundances are presented as molar ratios relative to Ca. The external reproducibility of the concentrations measured using this procedure is estimated at 2σ to be better than 15 % (RSD; $N = 9$) based on repeat measurements of an in-house standard solution.

For Ca isotopic analysis, samples were spiked by mixing a portion of samples, taken from the same solutions used for trace metal analysis, containing 7 μg of Ca with isotopically enriched ^{43}Ca - ^{42}Ca double spike in a 7:3 sample-to-spike ratio. The spiked samples were evaporated to dryness on a hotplate and brought up in 0.75 mL of 1.5N HCl prior to column chromatography to ensure sample-spike equilibration. Column chromatographic separation was performed using a set of gravity-flow chromatographic columns filled with MCI CK08P cation exchange resin following Wogsland (2020). Elution was done using distilled 1.5N HCl. The purified Ca solutions were refluxed for 24 h with hydrogen peroxide to remove residual organics before being evaporated to dryness on a hot plate. Sample preparation and column chromatography were performed in the School of Earth Sciences Class-100 clean room facility at Ohio State University.

3.2. Mass spectrometry

Calcium isotopic measurements were performed on a Thermo Scientific Triton Plus Thermal Ionization Mass Spectrometer (TIMS) at the TIMS Laboratory at Ohio State University following the procedure modified from Wogsland (2020) and briefly outlined here. Half of the prepared (spiked and purified) sample ($\sim 5 \mu\text{g}$) was dissolved in 1 μL of 1.5 N HCl and loaded onto hand-made outgassed single tantalum (Ta) filaments, with 1.25 μL of 3.2N ultrapure phosphoric acid as an activator. Isotopic measurements were done using a dynamic (“peak jumping”) collection routine with 3 cup configurations. Measurements were done in 10 blocks of 10 cycles. Potassium interference was monitored by measuring mass 41 in the center cup in the main cup configuration and confirmed that no correction was needed for ^{40}K interference on ^{40}Ca . The measured isotopic ratios are reported as the relative abundance of ^{44}Ca to ^{40}Ca in standard delta notation ($\delta^{44/40}\text{Ca}$) relative to modern seawater. Three samples of SRM915a and two samples of SRM915b were run within each sample wheel of up to 21 samples and yielded average values of $-1.88 \pm 0.08\text{‰}$ (2σ ; $N = 15$) and $-1.14 \pm 0.08\text{‰}$ (2σ ; $N = 10$), respectively, see Fig. S2 and Table S1. The analytical precision of this procedure is estimated at 0.08‰ based on repeat measurements of SRM 915a and 915b throughout the analytical period. Replicate measurements (average $2\sigma = 0.06\text{‰}$; $N = 9$) and a duplicate measurement (re-processing sample from start to finish, $2\sigma = 0.07\text{‰}$; $N = 1$) of the carbonate samples produce results that are within the analytical precision determined from repeat analysis of the carbonate standards SRM 915a and 915b.

4. Results

Chemostratigraphic profiles from Meiklejohn Peak are shown in Fig. 2 and tabulated in Table S2. The measured $\delta^{44/40}\text{Ca}$ values range from -0.91 to -1.38‰ (relative to seawater, Fig. 2). The base of the section is characterized by high $\delta^{44/40}\text{Ca}$ and low Sr/Ca values. The pre-MDICE $\delta^{13}\text{C}$ declining trend recorded within the Middle Member is also characterized by low Sr/Ca but a shift towards lower $\delta^{44/40}\text{Ca}$ values.

The peak of the MDICE $\delta^{13}\text{C}$ excursion is characterized by high Sr/Ca and low $\delta^{44/40}\text{Ca}$ values. The upper portion of the Upper Member is characterized by a decline in Sr/Ca and a shift toward higher $\delta^{44/40}\text{Ca}$ values. Cross plots between $\delta^{13}\text{C}$, $\delta^{44/40}\text{Ca}$, Sr/Ca (Fig. 3) and statistical analyses (Pearson, Spearman, and Kendall statistics) show that there is a significant covariation between Sr/Ca and $\delta^{44/40}\text{Ca}$, whereas $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$ show limited to no covariation (see supplementary materials Table S3).

5. Discussion

Our results from shallow marine carbonates at Meiklejohn Peak, Nevada reveal strong negative covariation in bulk carbonate Sr/Ca and $\delta^{44/40}\text{Ca}$, but a lack of significant covariation between $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$. Two factors that can produce negative covariation between Sr/Ca and $\delta^{44/40}\text{Ca}$ include mineralogy and diagenesis, which should both also influence $\delta^{13}\text{C}$. The following discussion first examines the evidence for changes in mineralogy and diagenesis that could have produced covariation between Sr/Ca - $\delta^{44/40}\text{Ca}$ and calculates the component of the $\delta^{13}\text{C}$ curve related to these processes. The residual $\delta^{13}\text{C}$ variability, which is decoupled from $\delta^{44/40}\text{Ca}$ and Sr/Ca, is assumed to be related to changes in the platform-scale water mass DIC or the global carbon cycle during the Middle Ordovician.

5.1. Variations in mineralogy

A first-order observation of our Meiklejohn Peak Ordovician data shows that the range of $\delta^{44/40}\text{Ca}$ falls between primary calcite and aragonite end members (see values in Gussone et al., 2005), suggesting mineralogy could have played a role in the observed stratigraphic variations because a change in the proportion of aragonite and calcite would also affect bulk carbonate $\delta^{13}\text{C}$ and Sr/Ca (Kimmig and Holmden, 2017). This notion is supported by the strong negative covariation between Sr/Ca vs $\delta^{44/40}\text{Ca}$, as primary aragonite is characterized by higher Sr concentrations and lower $\delta^{44/40}\text{Ca}$ values relative to primary calcite (Brand and Veizer, 1980; Gussone et al., 2005; Lau et al., 2017).

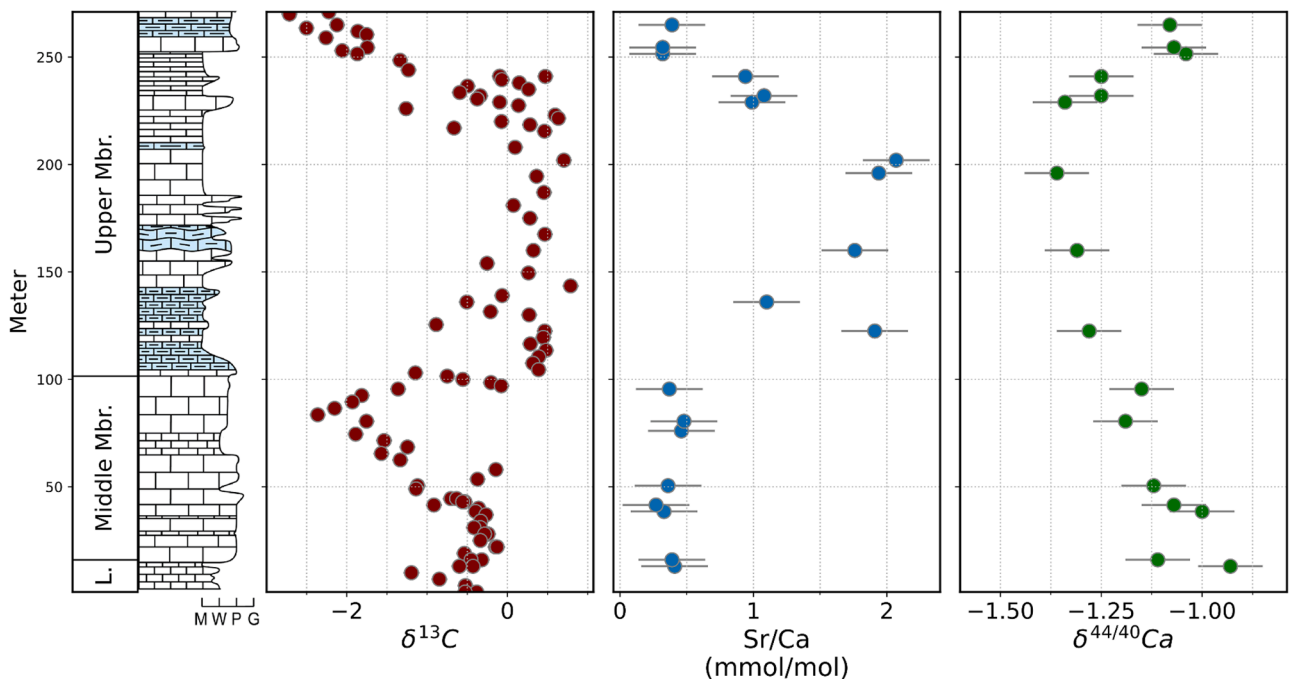


Fig. 2. Chemostratigraphic profile of the Middle Ordovician Meiklejohn Peak, Nevada section (see Fig. 1 for age constraints and lithologic details). Carbon isotope data ($\delta^{13}\text{C}$, ‰ V-PDB) are from Diamond (2021). Bulk carbonate Sr/Ca and $\delta^{44/40}\text{Ca}$ (relative to modern seawater) datasets are from this study. These datasets are generated from paired measurements of the same hand-specimens analyzed by Diamond (2021). Error bars for $\delta^{44/40}\text{Ca}$ are based on 2σ of repeat measurements of SRM 915a (0.08‰, $N = 15$). Lithostratigraphic unit abbreviations: L. = Lower Member of the Antelope Valley Formation; Mbr. = Member.

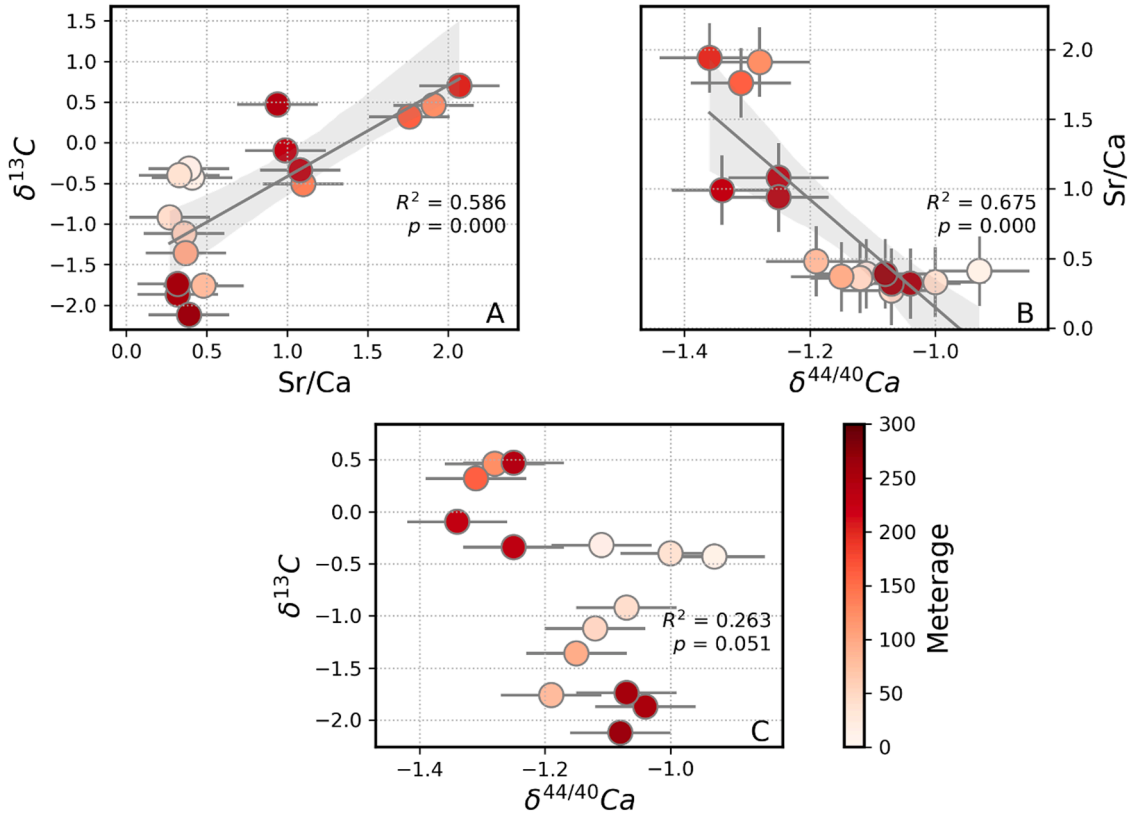


Fig. 3. Cross plots of bulk carbonate $\delta^{13}\text{C}$ vs Sr/Ca (A), Sr/Ca vs $\delta^{44/40}\text{Ca}$ (B), and $\delta^{13}\text{C}$ vs $\delta^{44/40}\text{Ca}$ (C), for Ordovician samples at Meiklejohn Peak, Nevada. The strength and statistical significance of covariation are presented as Pearson coefficient of determination R^2 and p -values, shown in each cross plot. The summary of all covariance analyses performed in this study is shown in Table S3 in the supplementary material. $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$ values are relative to V-PDB and modern seawater respectively. Sr/Ca values are in mmol/mol. The color scale reflects stratigraphic height (in meters).

Moreover, samples with low Sr/Ca ratios correspond to $\delta^{44/40}\text{Ca}$ values that are up to -0.9‰ which is close to what is expected for primary calcite (see Gussone et al., 2005).

We use a simple two-end member mixing model to quantify the potential influence of primary mineralogy on both $\delta^{44/40}\text{Ca}$ and $\delta^{13}\text{C}$ (Fig. 4). The model uses the mixing equation (Eq. (1)) from Faure and Mensing (2009) where R_x , f_x , C_x are isotopic ratios, fractions, and concentrations respectively. The subscripts correspond to bulk carbonate (carb), the aragonite end member (arag), and the calcite end member (calc). For the $\delta^{13}\text{C}$ mixing model, the isotopic ratio of each end member is calculated using isotopic offsets for aragonite (2.7‰) and calcite (1.0‰) fractionation from seawater in Romanek et al. (1992) and an assumed isotopic ratio ($\delta^{13}\text{C}$) of DIC of -3.7‰ . This DIC value is set to produce the pure calcite end member equal to the lightest $\delta^{13}\text{C}$ value from Meiklejohn Peak, which is -2.7‰ (i.e., isolating the mineralogic influence by assuming no role for diagenesis, which is discussed in the next section). For the $\delta^{44/40}\text{Ca}$ two component mixing model, the isotopic ratio of each end member is based on the isotopic offset of aragonite (-1.6‰) and calcite (-0.9‰) from Gussone et al. (2005) and seawater $\delta^{44/40}\text{Ca}$ of 0‰ .

$$R_{\text{carb}} = R_{\text{arag}} f_{\text{arag}} \frac{C_{\text{arag}}}{C_{\text{carb}}} + R_{\text{calc}} f_{\text{calc}} \frac{C_{\text{calc}}}{C_{\text{carb}}} \quad (1)$$

The mixing model shows that the range of measured $\delta^{44/40}\text{Ca}$ values falls within the theoretical mixing between aragonite and calcite, whereas the range of measured $\delta^{13}\text{C}$ covers a much wider range (Fig. 4). The mixing model suggests that variations in primary mineralogy could be a control on $\delta^{44/40}\text{Ca}$ (and Sr/Ca) at Meiklejohn Peak but cannot solely account for the $\delta^{13}\text{C}$ curve. Despite the lack of unequivocal evidence of aragonite deposition at Meiklejohn Peak (Ross et al., 1989), the low paleolatitude of the Meiklejohn Peak section (Fig. 1) and

paleotemperature reconstructions of the Middle Ordovician oceans (Grossman and Joachimski, 2020) suggest that both aragonite and calcite precipitation could occur in carbonate platform settings (Balthasar and Cusack, 2015). Furthermore, recent studies have found aragonitic components (shells and ooids) within Middle Ordovician carbonate strata in eastern Laurentia (e.g., James et al., 2020) despite this time period traditionally being interpreted as a calcite sea interval (Hardie, 1996). A prominent shift to higher Sr/Ca occurs at the boundary between the Middle and Upper Members of the Antelope Valley Limestone coincides with an overall increase in lime mud content (Fig. 2) that could have precipitated as aragonite, like the modern Bahamas (Geyman et al., 2022). If the calcite and aragonite minerals were precipitated from different DIC pools (open marine and platform top) with distinct $\delta^{13}\text{C}$ (like the Bahamas), the majority of the excursion may be explained by a simple mixing in calcite and aragonite (see Fig. S3). It is worth noting that the depositional setting of the carbonate rocks exposed at Meiklejohn Peak is thought to be a ramp carbonate platform (Ross et al., 1989) and may not be analogous to the Bahamas (isolated carbonate platform).

If we assume that our samples do not reflect precipitation from spatially heterogeneous platform top DIC pool, the wide range of $\delta^{13}\text{C}$ values that fall beyond the theoretical aragonite-calcite mixing line (Fig. 4) suggests that variations in the proportion of aragonite and calcite cannot be the sole cause of the observed $\delta^{13}\text{C}$ curve (Fig. 5C). This suggests that the $\delta^{13}\text{C}$ composition of DIC would have had to evolve (locally or globally) independent of any changes in primary carbonate mineralogy during the Ordovician to reproduce the $\delta^{13}\text{C}$ curve at Meiklejohn Peak or that diagenesis played a role. It is noteworthy that the 1.7‰ offset between aragonite and calcite is based on a laboratory experiment (Romanek et al., 1992) and carbonate precipitation in natural settings may produce isotopic offsets that are either higher or lower

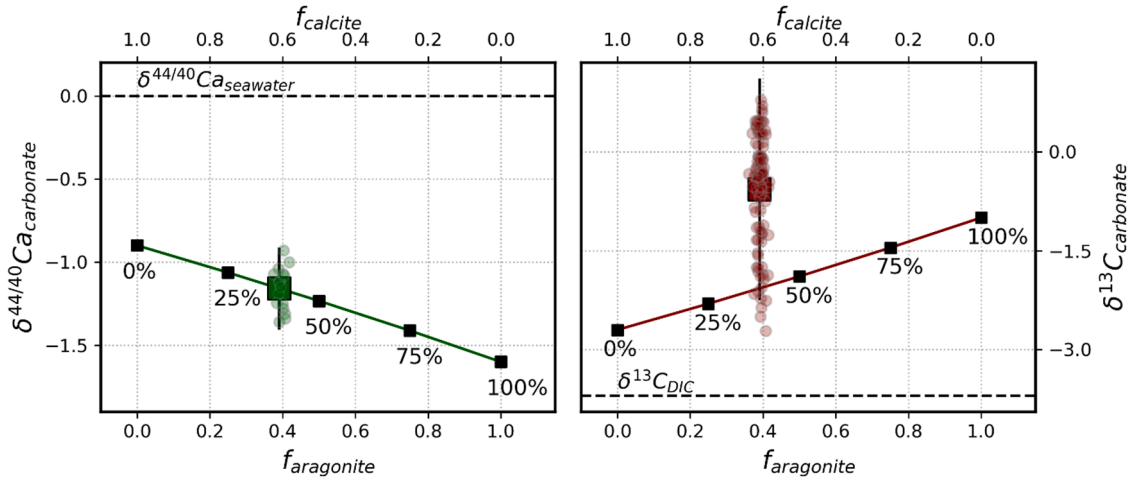


Fig. 4. Simple two-end member mixing model of aragonite and calcite. The x-axes represent the fraction of each end member (lower axes: aragonite; upper axes: calcite). The boxes and whiskers in each plot represent the range of measured geochemical values from Meiklejohn Peak (mean $\pm 2\sigma$). The box and whisker plots are tied to the y-axes. The placement of the boxes on x-axes represents the estimated fraction of aragonite ($f_{\text{aragonite}}$) based on the mean value of $\delta^{44/40}\text{Ca}$ ($\sim 40\%$). The range of values for $\delta^{44/40}\text{Ca}$ falls within the range of mixing lines, whereas the range of values for $\delta^{13}\text{C}$ is roughly twice the range of the calcite-aragonite mixing line. The $\delta^{13}\text{C}$ data points that fall outside of the mixing line ($> 100\%$ aragonite) suggest that variations in the carbonate mineralogy alone ($f_{\text{aragonite}}$ vs f_{calcite}) could not have fully controlled the observed $\delta^{13}\text{C}$ variations.

than this accepted value due to possible vital effects and physiological microenvironments (Romanek et al., 1992). Nevertheless, if we assume mineralogy (and not diagenesis) was the main control of $\delta^{44/40}\text{Ca}$, then we can estimate the required change in $\delta^{13}\text{C}$ composition of DIC in our record using Eq. (2) (Fig. 5D). This approach suggests that the largest observed shift in the bulk $\delta^{13}\text{C}$ ($\sim +3\%$) would have required a $+1.5\%$ shift in platform water mass and/or global ocean DIC. The remaining $+1.5\%$ shift is due to isotopic offset associated with varying proportions of carbonate polymorphs (aragonite + calcite).

$$\text{DIC} = \delta^{13}\text{C}_{\text{carb}} - (\Delta_{\text{arag}}f_{\text{arag}} + \Delta_{\text{calc}}f_{\text{calc}}) \quad (2)$$

5.2. Early marine diagenesis and kinetics

5.2.1. Early marine diagenesis and modeling

An alternative process that can explain the observed negative covariation between Sr/Ca and $\delta^{44/40}\text{Ca}$ is early marine diagenesis, which should also impact $\delta^{13}\text{C}$. Dissolution and reprecipitation during early marine diagenesis tend to produce heavier $\delta^{44/40}\text{Ca}$ values due to equilibration with seawater (e.g., Fantle and DePaolo, 2007; Higgins et al., 2018; Jacobson and Holmden, 2008) which drives down $\delta^{13}\text{C}$ if the style of diagenesis is fluid-buffered by seawater (Ahm et al., 2018), also lowering Sr/Ca ratios (Brand and Veizer, 1980; Fantle et al., 2020). Recent studies on modern and ancient carbonate platforms suggest that the style of diagenesis can switch between being buffered by the surrounding rock (sediment-buffered) or the diagenetic fluid (fluid-buffered) during sea level change (Ahm et al., 2018; Higgins et al., 2018; Lau and Hardisty, 2022). Using this interpretive framework in our Middle Ordovician sediments at Meiklejohn Peak, the Lower Member and the uppermost portion of the Upper Member of the Antelope Valley Limestone that have high $\delta^{44/40}\text{Ca}$ and low Sr/Ca values can be interpreted to have been altered during fluid-buffered diagenesis, whereas most of the Upper Member (lowest $\delta^{44/40}\text{Ca}$ and highest Sr/Ca) was sediment-buffered during early marine diagenesis. As was the case with changes in mineralogy discussed above, this diagenetic scenario can be used to explain some but not all of the changes in $\delta^{13}\text{C}$ with the portion that is decoupled from $\delta^{44/40}\text{Ca}$ and Sr/Ca likely due to changes in the $\delta^{13}\text{C}$ of DIC (locally or globally).

To explore the influence of early marine diagenesis on $\delta^{44/40}\text{Ca}$, Sr/Ca, and $\delta^{13}\text{C}$ we apply the quantitative modeling framework developed by Ahm et al. (2018) to our Meiklejohn Peak data. This model simulates

recrystallization and/or neomorphism processes during early marine diagenesis. In this modeling framework, neomorphism is strictly defined as a transformation from aragonite to calcite through recrystallization. In this modeling scenario, we assume that unaltered rocks consist of 100 % aragonite (primary mineral) and early marine diagenesis transforms the primary mineral into calcite (diagenetic mineral). These minerals are defined stoichiometrically (i.e., the difference between aragonite and calcite is their major (Ca) and trace element (Sr) concentrations). One of the main assumptions of the Ahm et al. (2018) model is that the entire solid phase reaches equilibrium at each modeling step which may not always be the case (e.g., Hashim and Kaczmarek, 2021). We used the default parameters of Ahm et al. (2018), except for the $\delta^{13}\text{C}$ of the diagenetic fluid (seawater) that we set to -3.7% , which is much lighter than modern carbonate platforms such as the Bahamas but plausible based on Panchuk et al. (2006) whose modeling scenarios recognize the influence of terrestrial organic matter input on the Ordovician epicontinental sea environment in Laurentia. We use -3.7% as the lightest possible $\delta^{13}\text{C}$ value for seawater from Panchuk et al. (2006) that can produce the lightest $\delta^{13}\text{C}$ value at Meiklejohn Peak of -2.7% (if calcite precipitation is assumed). The $\delta^{13}\text{C}$ of primary sediment is set to 0.7% , which is based on the assumption that the sample with the highest Sr/Ca of 2.04 (Table S2) is primary (see Fig. S4 for sensitivity analysis of the primary sediment $\delta^{13}\text{C}$ value). This primary $\delta^{13}\text{C}$ is much lighter than modern carbonate in the Bahamas, but possible based on the significantly lighter DIC in parts of the Ordovician epicontinental seas (Panchuk et al., 2006). The $\delta^{44/40}\text{Ca}$ of the diagenetic fluid is set to -0.5% to accommodate uncertainty associated with Ca isotopic fractionations during diagenetic recrystallization and possible influx of additional light $\delta^{44/40}\text{Ca}$ from upstream carbonate dissolution (Ahm et al., 2018; Higgins et al., 2018). The full set of parameters are presented in Table 1.

The results of the diagenetic modeling are shown in Fig. 6 and suggest that early marine diagenesis cannot reproduce the range of $\delta^{13}\text{C}_{\text{carb}}$ values from Meiklejohn Peak, consistent with the less significant covariation between $\delta^{13}\text{C}$ vs $\delta^{44/40}\text{Ca}$ ($p\text{-value} > 0.05$; Fig. 3). If the numerical model accurately captures the isotopic mass balance during early marine diagenesis, diagenetic alteration may account for $\sim 50\%$ of the range of observed $\delta^{13}\text{C}$ at Meiklejohn Peak (from ~ 0.5 to -1% that falls within the diagenetic envelope). However, as summarized by Crockford et al. (2020), some departures of data points from the predicted model solution using the Ahm et al. (2018) framework may be expected because of heterogeneities in the primary sediment or the

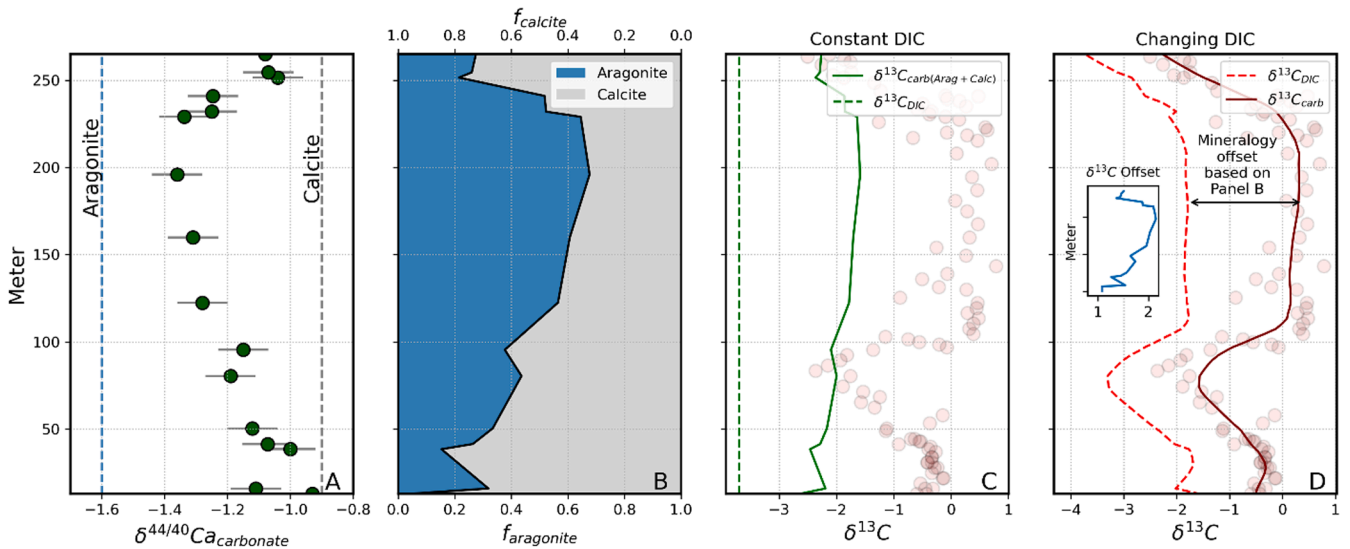


Fig. 5. Estimated $\delta^{13}\text{C}$ of DIC and carbonate sediment based on modeled mineralogical variations at Meiklejohn Peak from calcium isotopes. (A) Calcium isotopes ($\delta^{44/40}\text{Ca}$) and the isotopic ratios of the assumed aragonite and calcite end members (R_{arag} and R_{calc} of Eq. (1)) used to reconstruct the primary mineral composition ($f_{\text{aragonite}}$ and f_{calcite}) shown in panel B. (B) Modeled primary mineral composition from calcium isotopes based on Eq. (1). (C) Estimated $\delta^{13}\text{C}_{\text{carb}}$ (solid green line) from fixed DIC ($\delta^{13}\text{C}_{\text{DIC}}$) (dashed green line) assuming mineralogy from $\delta^{44/40}\text{Ca}$ (panel B). (D) Carbonate carbon isotope LOESS curve ($\delta^{13}\text{C}_{\text{carb}}$) of Middle Ordovician samples at Meiklejohn Peak (solid red line) and the estimated DIC $\delta^{13}\text{C}$ (dashed red line) based on the modeled mineralogical composition from $\delta^{44/40}\text{Ca}$ (panel B) using Eq. (2). This model assumes that primary mineral composition is the sole factor affecting $\delta^{44/40}\text{Ca}$ and $\delta^{13}\text{C}_{\text{carb}}$. Inset shows net $\delta^{13}\text{C}_{\text{carb}}$ offset applied to estimate DIC $\delta^{13}\text{C}$ from measured $\delta^{13}\text{C}_{\text{carb}}$.

diagenetic fluid (e.g., due to inputs of remineralized organic matter that affect $\delta^{13}\text{C}$ and not $\delta^{44/40}\text{Ca}$).

To fit all the $\delta^{13}\text{C}$ data from Meiklejohn Peak in the model space requires an extremely light $\delta^{13}\text{C}$ isotopic ratio of the diagenetic fluid ($\sim -10\text{‰}$) (Fig. 6 inset). While it is impossible to have a global DIC value of -10‰ during the Phanerozoic (Kump and Arthur, 1999; Planavsky et al., 2022), there are several mechanisms that can drive $\delta^{13}\text{C}$ of local DIC toward lighter values and decouple from $\delta^{44/40}\text{Ca}$ (e.g., Crockford et al., 2021). Patterson and Walter (1994) suggested that the influx of organic carbon from terrestrial environments and its subsequent oxidation drive the DIC in Florida Bay toward lighter $\delta^{13}\text{C}$ ($\sim -7\text{‰}$) relative to open-ocean water ($1.5 - 2.0\text{‰}$). However, although terrestrial input of organic carbon may have played a role in the Ordovician epeiric sea $\delta^{13}\text{C}$ of DIC that could be as light as -4 to -5‰ (Panchuk et al., 2006), values as light as -10‰ are unlikely during the Middle Ordovician because of the absence of well-established terrestrial ecosystems (Jones et al., 2015; Morris et al., 2018).

In addition to the variables discussed above that can influence the isotopic composition of the diagenetic fluid, both spatially and temporally, the composition of the primary carbonate can also be influenced by platform-scale processes. One mechanism that can drive shallow platform DIC and precipitated carbonate toward lighter $\delta^{13}\text{C}$ values is atmospheric CO_2 invasion, which has been suggested to play an important role in the extremely light $\delta^{13}\text{C}$ values observed during the Neoproterozoic (Ahm et al., 2021; Busch et al., 2022). Carbon dioxide invasion from the atmosphere to a local DIC pool can induce a strong kinetic fractionation that depletes $\delta^{13}\text{C}$ down to -11‰ (Lazar and Erez, 1990) and is associated with the hydroxylation of aqueous CO_2 (Clark et al., 1992). Carbon isotopic depletion through atmospheric CO_2 invasion requires a disequilibrium condition between the DIC pool and the atmosphere, which can be achieved in highly evaporative conditions (Lazar and Erez, 1990). However, while parts of the Antelope Valley Limestone exposed at Meiklejohn Peak are thought to represent shallow carbonate platform environments (Fig. 1; Ross et al. (1989)), the lack of sedimentologic evidence of an intense evaporative condition combined with the presence of nektonic open marine fossils (i.e., conodonts) together suggest that atmospheric CO_2 invasion due to high evaporation was unlikely.

5.2.2. Assessment of potential role for precipitation rates and kinetics

Alternatively, kinetic effects due to variations in the carbonate precipitation rates irrespective of a change in mineralogy could also have played a role in causing the observed stratigraphic variations in $\delta^{44/40}\text{Ca}$ and Sr/Ca at Meiklejohn Peak, as supported by Sr/Ca and $\delta^{44/40}\text{Ca}$ negative covariation (Wang et al., 2023, 2021), which could also potentially affect $\delta^{13}\text{C}$. However, a comparison between our dataset to an experimental study by Tang et al. (2008a) (Fig. S5) shows very different slopes between $\delta^{44/40}\text{Ca}$ and Sr/Ca suggesting that kinetic effects are unlikely to be the main process that affects the observed $\delta^{44/40}\text{Ca}$ and Sr/Ca from Meiklejohn Peak. Moreover, kinetic effects should normally produce a positive covariation between $\delta^{44/40}\text{Ca}$ and $\delta^{13}\text{C}$ which we do not observe in this section. A future study that pairs $\delta^{44/40}\text{Ca}$ and stable strontium ($\delta^{88/86}\text{Sr}$) could be used to further constrain the role of kinetic effects (e.g., Wang et al., 2023).

5.3. A role for changing DIC of the platform water mass and/or global ocean

Because $\delta^{44/40}\text{Ca}$ and $\delta^{13}\text{C}$ were at least partially decoupled in our Meiklejohn Peak dataset (Fig. 3), changes in local or global marine DIC are a likely mechanism for the portion of the trend in $\delta^{13}\text{C}$ that cannot be accounted for by either mineralogy or diagenesis. Changes in local DIC have been invoked as drivers for numerous $\delta^{13}\text{C}$ excursions during the Precambrian including the Trezona (Ahm et al., 2021) and Shuram excursions (Busch et al., 2022). In both scenarios, they argue that changes in local DIC associated with a global transgression paired with enhanced evaporation and primary productivity (Ahm et al., 2021; Busch et al., 2022) were the most likely driver of stratigraphic $\delta^{13}\text{C}$ excursions, which could also be modified spatially across the platform by diagenesis. This scenario, however, is unlikely for our data from Meiklejohn Peak because, there is no evidence for highly evaporative conditions which is the requirement for CO_2 invasion (Lazar and Erez, 1990) or non-equilibrium gas transfer isotopic fractionation (Stiller et al., 1985) that can drive large shifts in local DIC. Furthermore, in both Trezona and Shuram models, local changes in the DIC are supported by strong facies-dependent variations in both $\delta^{13}\text{C}$ and $\delta^{44/40}\text{Ca}$ which we do not observe at Meiklejohn Peak (see Fig. S6 in the supplementary materials).

Table 1

Early marine diagenesis modeling parameters. Table includes justification for parameter selections if different from the default values used in [Ahm et al. \(2018\)](#).

Parameters	Definition	Values	Justification
α_{Ca}	Ca isotopic fractionation factor for diagenesis	1.000	
α_C	C isotopic fractionation factor for diagenesis	1.001	
K_{sr}	Distribution coefficient for Sr	0.05	
R	Reaction rate	10 %/myr	
u	Advection rate	10 cm/yr	
Diagenetic mineral		Calcite	
Primary mineral		Aragonite	
M_f	Mass of element in diagenetic fluid (mol/kg)	C 0.0024	
		Ca 0.035	
		Sr 0.00035	
M_s	Mass of element in primary sediment (ppm)	C 120000	
		Ca 388500	
		Sr 9000	
d_f	Isotopic value of fluid	$\delta^{13}C$ -3.7	DIC to produce the lightest $\delta^{13}C$
		$\delta^{44}Ca$ -0.5	
d_s	Isotopic value of primary sediment	$\delta^{13}C$ 0.7	$\delta^{13}C$ of sample with the highest Sr/Ca
		$\delta^{44}Ca$ -1.6	

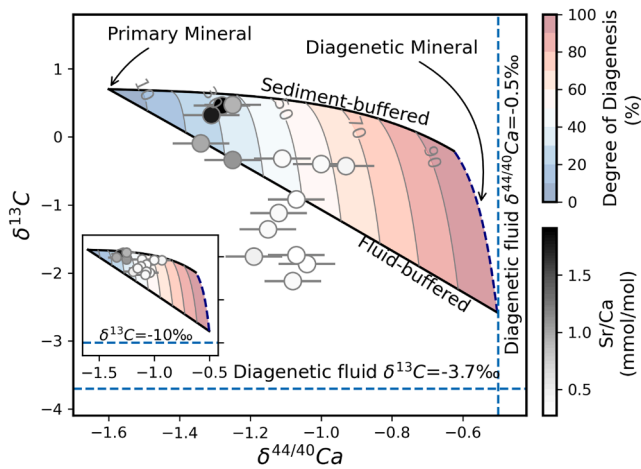


Fig. 6. Early marine diagenesis model based on [Ahm et al. \(2018\)](#). The thick black lines represent the diagenetic trajectory of the fluid-buffered and sediment-buffered end members (sensu [Higgins et al., 2018](#)). The contour lines show the degree of diagenesis (i.e., the proportion of diagenetic minerals relative to primary minerals). The dashed black line represents the diagenetic mineral end member. The primary mineral and diagenetic mineral are set to aragonite and calcite respectively. The $\delta^{13}C$ of diagenetic fluid is set at -3.7‰, whereas $\delta^{44}Ca$ is set at -0.5‰ following [Ahm et al \(2018\)](#) (marked by blue dashed lines). The inset shows a modeling scenario that fits all data, which requires the $\delta^{13}C$ of diagenetic fluid to be set at -10‰. This extremely light fluid value is highly unlikely given the open marine facies of the Meiklejohn Peak section.

The lack of facies-dependent variations in $\delta^{44}Ca$ also suggests that the stratigraphic variations in $\delta^{44}Ca$ observed at Meiklejohn Peak are unlikely to be a result of changes in carbonate platform hydrodynamics related to shifts in the depositional environment. Nevertheless, it is worth noting that a previous study suggests that local changes in DIC can amplify the global DIC signal ([Saltzman and Edwards, 2017](#)).

While we cannot completely rule out possible changes in local DIC at Meiklejohn Peak, an alternative explanation for the $\delta^{13}C$ shift is a change in global DIC due to increased organic carbon burial associated with enhanced primary productivity and increasing nutrient delivery to the ocean. Previous modeling studies have suggested that marine phosphate is the ultimate limiting nutrient throughout Earth history (e. g., [Laakso and Schrag, 2014](#)), and because continental weathering is the main source of phosphate to the modern oceans ([Ruttenberg, 2003](#)), an increase in continental weathering could potentially drive increased nutrient delivery. The Middle Ordovician Period recorded by our Meiklejohn Peak dataset broadly coincides with a possible increase in basaltic weathering associated with the Taconic uplift ([Conwell et al., 2022](#); [Saltzman and Edwards, 2017](#); [Swanson-Hysell and Macdonald, 2017](#); [Wu et al., 2017](#)). Therefore, enhanced basaltic weathering associated with the Taconic uplift may have led to an increase in nutrient delivery and primary productivity which eventually caused a positive shift in the $\delta^{13}C$. Further, this process may have a direct link to environmental changes during the Middle Ordovician which set the stage for a rapid increase in marine biodiversity known as the Great Ordovician Biodiversification Event ([Rasmussen et al., 2016](#); [Stigall et al., 2019](#)).

6. Conclusion

We present paired measurements of $\delta^{13}C$, $\delta^{44}Ca$, and Sr/Ca from an Ordovician stratigraphic section at Meiklejohn Peak in Nevada. The data and numerical models presented here suggest that variations in Sr/Ca and $\delta^{44}Ca$ from Meiklejohn Peak can be explained by early marine diagenesis and/or changes in the dominant carbonate primary mineralogy. However, the $\delta^{13}C$ values at least in part likely reflect a primary change in the local or global DIC. If the excursion is a reflection of a shift in the global DIC, the $\delta^{13}C$ excursion recorded at Meiklejohn Peak may represent a transient increase in primary productivity and subsequent organic carbon burial during the Ordovician due to an increase in nutrient delivery associated with basaltic weathering during the Taconic uplift.

CRedit authorship contribution statement

Y. Datu Adiatma: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Matthew R. Saltzman:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Elizabeth M. Griffith:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and python codes related to this article are available online at <https://github.com/datuadiatma/CaOrdovician>.

Acknowledgements

This work was financially supported in part by the School of Earth Sciences, Friends of Orton Hall student research funds to YDA, and National Science Foundation Award EAR-2221962 to MRS and EMG. We thank Matt Braun, Peter Haber, and Cole Edwards for helpful discussions. This manuscript benefits from comments and feedback by two anonymous reviewers and editorial guidance by Andrew Jacobson.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.epsl.2024.118805](https://doi.org/10.1016/j.epsl.2024.118805).

References

- Ahm, A.S.C., Bjerrum, C.J., Blättler, C.L., Swart, P.K., Higgins, J.A., 2018. Quantifying early marine diagenesis in shallow-water carbonate sediments. *Geochim. Cosmochim. Acta* 236, 140–159. <https://doi.org/10.1016/j.gca.2018.02.042>.
- Ahm, A.S.C., Bjerrum, C.J., Hoffman, P.F., Macdonald, F.A., Maloof, A.C., Rose, C.V., Strauss, J.V., Higgins, J.A., 2021. The Ca and Mg isotope record of the Cryogenian Trezona carbon isotope excursion. *Earth Planet. Sci. Lett.* 568, 117002 <https://doi.org/10.1016/j.epsl.2021.117002>.
- Ainsaar, L., Kaljo, D., Martma, T., Meidla, T., Männik, P., Nõlvak, J., Tinn, O., 2010. Middle and Upper Ordovician carbon isotope chemostratigraphy in Baltoscandia: a correlation standard and clues to environmental history. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 294, 189–201. <https://doi.org/10.1016/j.palaeo.2010.01.003>.
- Albanesi, G.L., Bergström, S.M., Schmitz, B., Serra, F., Feltes, N.A., Voldman, G.G., Ortega, G., 2013. Darrivilian (Middle Ordovician) $\delta^{13}\text{C}_{\text{carb}}$ chemostratigraphy in the Precordillera de Argentina: documentation of the middle Darrivilian Isotope Carbon Excursion (MDICE) and its use for intercontinental correlation. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 389, 48–63. <https://doi.org/10.1016/j.palaeo.2013.02.028>. SI: Pre-Mesozoic climates and global change.
- Allan, J.R., Matthews, R.K., 1982. Isotope signatures associated with early meteoric diagenesis. *Sedimentology* 29, 797–817. <https://doi.org/10.1111/j.1365-3091.1982.tb00085.x>.
- Balthasar, U., Cusack, M., 2015. Aragonite-calcite seas—quantifying the gray area. *Geology* 43, 99–102. <https://doi.org/10.1130/G36293.1>.
- Berner, R.A., 2006. GEOCARBSULF: a combined model for Phanerozoic atmospheric O_2 and CO_2 . *Geochim. Cosmochim. Acta* 70, 5653–5664. <https://doi.org/10.1016/j.gca.2005.11.032>. A Special Issue Dedicated to Robert A. Berner.
- Brand, U., Veizer, J., 1980. Chemical diagenesis of a multicomponent carbonate system; 1, Trace elements. *J. Sediment. Res.* 50, 1219–1236. <https://doi.org/10.1306/212F7B87-2B24-11D7-8648000102C1865D>.
- Busch, J.F., Hodgins, E.B., Ahm, A.S.C., Huxson, J.M., Macdonald, F.A., Bergmann, K.D., Higgins, J.A., Strauss, J.V., 2022. Global and local drivers of the Ediacaran Shuram carbon isotope excursion. *Earth Planet. Sci. Lett.* 579, 117368 <https://doi.org/10.1016/j.epsl.2022.117368>.
- Clark, I.D., Fontes, J.C., Fritz, P., 1992. Stable isotope disequilibria in travertine from high pH waters: laboratory investigations and field observations from Oman. *Geochim. Cosmochim. Acta* 56, 2041–2050. [https://doi.org/10.1016/0016-7037\(92\)90328-G](https://doi.org/10.1016/0016-7037(92)90328-G).
- Conwell, C.T., Saltzman, M.R., Edwards, C.T., Griffith, E.M., Adiatma, Y.D., 2022. Nd isotope evidence for enhanced mafic weathering leading to Ordovician cooling. *Geology* 50, 886–890. <https://doi.org/10.1130/G49860.1>.
- Cramer, B.D., Jarvis, I., 2020. Chapter 11 - carbon isotope stratigraphy. Gradstein, F.M., Ogg, J.G., Schmitz, M.D., Ogg, G.M. (Eds.). *Geologic Time Scale 2020*. Elsevier, pp. 309–343. <https://doi.org/10.1016/B978-0-12-824360-2.00011-5>.
- Crockford, P.W., Kunzmann, M., Blättler, C.L., Kalderon-Asael, B., Murphy, J.G., Ahm, A.S., Sharoni, S., Halverson, G.P., Planavsky, N.J., Halevy, I., Higgins, J.A., 2021. Reconstructing Neoproterozoic seawater chemistry from early diagenetic dolomite. *Geology* 49, 442–446. <https://doi.org/10.1130/G48213.1>.
- Diamond, C.W., 2021. *Animals, Oxygen, and the Mid-Proterozoic Earth System* (Ph.D.). University of California, Riverside, United States – California.
- Edwards, C.T., Saltzman, M.R., 2016. Paired carbon isotopic analysis of Ordovician bulk carbonate ($\delta^{13}\text{C}_{\text{carb}}$) and organic matter ($\delta^{13}\text{C}_{\text{org}}$) spanning the Great Ordovician Biodiversification Event. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 458, 102–117. <https://doi.org/10.1016/j.palaeo.2015.08.005>. The Ordovician Revolution: Co-Evolution of Climate and the Biosphere.
- Edwards, C.T., Saltzman, M.R., Leslie, S.A., Bergström, S.M., Sedlacek, A.R.C., Howard, A., Bauer, J.A., Sweet, W.C., Young, S.A., 2015. Strontium isotope ($^{87}\text{Sr}/^{86}\text{Sr}$) stratigraphy of Ordovician bulk carbonate: implications for preservation of primary seawater values. *Geol. Soc. Am. Bull.* 127, 1275–1289. <https://doi.org/10.1130/B31149.1>.
- Edwards, C.T., Saltzman, M.R., Royer, D.L., Fike, D.A., 2017. Oxygenation as a driver of the great ordovician biodiversification event. *Nat. Geosci.* 10, 925–929. <https://doi.org/10.1038/s41561-017-0006-3>.
- Fantle, M.S., Barnes, B.D., Lau, K.V., 2020. The Role of diagenesis in shaping the geochemistry of the marine carbonate record. *Annu. Rev. Earth Planet. Sci.* 48, 549–583. <https://doi.org/10.1146/annurev-earth-073019-060021>.
- Fantle, M.S., DePaolo, D.J., 2007. Ca isotopes in carbonate sediment and pore fluid from ODP Site 807A: The $\text{Ca}^{2+}(\text{aq})$ -calcite equilibrium fractionation factor and calcite recrystallization rates in Pleistocene sediments. *Geochim. Cosmochim. Acta* 71, 2524–2546. <https://doi.org/10.1016/j.gca.2007.03.006>.
- Faure, G., Mensing, T.M., 2009. *Isotopes Principles and Applications*, 3rd ed. Wiley.
- Geyman, E.C., Maloof, A.C., 2021. Facies control on carbonate $\delta^{13}\text{C}$ on the Great Bahama Bank. *Geology* 49, 1049–1054. <https://doi.org/10.1130/G48862.1>.
- Geyman, E.C., Maloof, A.C., 2019. A diurnal carbon engine explains ^{13}C -enriched carbonates without increasing the global production of oxygen. *Proc. Natl. Acad. Sci.* 116, 24433–24439. <https://doi.org/10.1073/pnas.1908783116>.
- Geyman, E.C., Wu, Z., Nadeau, M.D., Edmonds, S., Turner, A., Purkis, S.J., Howes, B., Dyer, B., Ahm, A.S.C., Yao, N., Deutsch, C.A., Higgins, J.A., Stolper, D.A., Maloof, A.C., 2022. The origin of carbonate mud and implications for global climate. *Proc. Natl. Acad. Sci.* 119, e2210617119 <https://doi.org/10.1073/pnas.2210617119>.
- Gischler, E., Swart, P.K., Lomando, A.J., 2009. Stable Isotopes of carbon and oxygen in modern sediments of carbonate platforms, barrier reefs, atolls and ramps: patterns and implications, in: perspectives in carbonate geology. John Wiley & Sons, Ltd, pp. 61–74. <https://doi.org/10.1002/9781444312065.ch5>.
- Goldman, D., Sadler, P.M., Leslie, S.A., Melchior, M.J., Agterberg, F.P., Gradstein, F.M., 2020. Chapter 20 - the Ordovician period. Gradstein, F.M., Ogg, J.G., Schmitz, M.D., Ogg, G.M. (Eds.). *Geologic Time Scale 2020*. Elsevier, pp. 631–694. <https://doi.org/10.1016/B978-0-12-824360-2.00020-6>.
- Griffith, E.M., Fantle, M.S., 2020. Calcium isotopes. *Elem. Geochem. Tracers Earth Syst. Sci.* <https://doi.org/10.1017/9781108853972>.
- Grossman, E.L., Joachimski, M.M., 2020. Chapter 10 - oxygen isotope stratigraphy. Gradstein, F.M., Ogg, J.G., Schmitz, M.D., Ogg, G.M. (Eds.). *Geologic Time Scale 2020*. Elsevier, pp. 279–307. <https://doi.org/10.1016/B978-0-12-824360-2.00010-3>.
- Gussone, N., Böhm, F., Eisenhauer, A., Dietzel, M., Heuser, A., Teichert, B.M.A., Reitner, J., Wörheide, G., Dullo, W.C., 2005. Calcium isotope fractionation in calcite and aragonite. *Geochim. Cosmochim. Acta* 69, 4485–4494. <https://doi.org/10.1016/j.gca.2005.06.003>.
- Hardie, L.A., 1996. Secular variation in seawater chemistry: An explanation for the coupled secular variation in the mineralogies of marine limestones and potash evaporites over the past 600 m.y. *Geology* 24, 279–283. [https://doi.org/10.1130/0091-7613\(1996\)024<0279:SVISCA>2.3.CO;2](https://doi.org/10.1130/0091-7613(1996)024<0279:SVISCA>2.3.CO;2).
- Harris, A.G., Bergström, S.M., Ethington, R.L., Ross Jr, R.J., 1979. Aspects of Middle and Upper Ordovician Conodont Biostratigraphy of Carbonate Facies in Nevada and Southeast California and Comparison with Some Appalachian Successions. *Brigham Young University Geology Studies*, pp. 7–44. Hamblin, W.K., Gardner, C.M. (Eds.).
- Hashim, M.S., Kaczmarek, S.E., 2021. The transformation of aragonite to calcite in the presence of magnesium: implications for marine diagenesis. *Earth Planet. Sci. Lett.* 574, 117166 <https://doi.org/10.1016/j.epsl.2021.117166>.
- Higgins, J.A., Blättler, C.L., Lundstrom, E.A., Santiago-Ramos, D.P., Akhtar, A.A., Crüger, A.M., Bialik, O., Holmden, C., Bradbury, H., Murray, S.T., Swart, P.K., 2018. Mineralogy, early marine diagenesis, and the chemistry of shallow-water carbonate sediments. *Geochim. Cosmochim. Acta* 220, 512–534. <https://doi.org/10.1016/j.gca.2017.09.046>.
- Husson, J.M., Higgins, J.A., Maloof, A.C., Schoene, B., 2015. Ca and Mg isotope constraints on the origin of Earth's deepest $\delta^{13}\text{C}$ excursion. *Geochim. Cosmochim. Acta* 160, 243–266. <https://doi.org/10.1016/j.gca.2015.03.012>.
- Husson, J.M., Schoene, B., Blüher, S., Maloof, A.C., 2016. Chemostratigraphic and U-Pb geochronologic constraints on carbon cycling across the Silurian–Devonian boundary. *Earth Planet. Sci. Lett.* 436, 108–120. <https://doi.org/10.1016/j.epsl.2015.11.044>.
- Jacobson, A.D., Holmden, C., 2008. ^{84}Ca evolution in a carbonate aquifer and its bearing on the equilibrium isotope fractionation factor for calcite. *Earth Planet. Sci. Lett.* 270, 349–353. <https://doi.org/10.1016/j.epsl.2008.03.039>.
- James, N.P., Narbonne, G.M., Armstrong, A.K.R., 2020. Aragonite depositional facies in a Late Ordovician calcite sea, Eastern Laurentia. *Sedimentology* 67, 3513–3532. <https://doi.org/10.1111/sed.12753>.
- Jones, D.S., Creel, R.C., Rios, B., Santiago Ramos, D.P., 2015. Chemostratigraphy of an Ordovician–Silurian carbonate platform: $\delta^{13}\text{C}$ records below glacioeustatic exposure surfaces. *Geology* 43, 59–62. <https://doi.org/10.1130/G36236.1>.
- Katz, M.E., Wright, J.D., Miller, K.G., Cramer, B.S., Fennel, K., Falkowski, P.G., 2005. Biological overprint of the geological carbon cycle. *Mar. Geol.* 217, 323–338. <https://doi.org/10.1016/j.margeo.2004.08.005>. Ocean Chemistry over the Phanerozoic and its links to Geological Processes.
- Kimmig, S.R., Holmden, C., 2017. Multi-proxy geochemical evidence for primary aragonite precipitation in a tropical-shelf ‘calcite sea’ during the Hirnantian glaciation. *Geochim. Cosmochim. Acta* 206, 254–272. <https://doi.org/10.1016/j.gca.2017.03.010>.
- Kinsman, D.J.J., 1969. Interpretation of Sr (super +2) concentrations in carbonate minerals and rocks. *J. Sediment. Res.* 39, 486–508. <https://doi.org/10.1306/74D71CB7-2B21-11D7-8648000102C1865D>.
- Knoll, A.H., Hayes, J.M., Kaufman, A.J., Swett, K., Lambert, I.B., 1986. Secular variation in carbon isotope ratios from Upper Proterozoic successions of Svalbard and East Greenland. *Nature* 321, 832–838. <https://doi.org/10.1038/321832a0>.
- Kump, L.R., Arthur, M.A., 1999. Interpreting carbon-isotope excursions: carbonates and organic matter. *Chem. Geol.* 161, 181–198. [https://doi.org/10.1016/S0009-2541\(99\)00086-8](https://doi.org/10.1016/S0009-2541(99)00086-8).
- Laakso, T.A., Schrag, D.P., 2014. Regulation of atmospheric oxygen during the Proterozoic. *Earth Planet. Sci. Lett.* 388, 81–91. <https://doi.org/10.1016/j.epsl.2013.11.049>.

- Lau, K.V., Hardisty, D.S., 2022. Modeling the impacts of diagenesis on carbonate paleoredox proxies. *Geochim. Cosmochim. Acta* 337, 123–139. <https://doi.org/10.1016/j.gca.2022.09.021>.
- Lau, K.V., Maher, K., Brown, S.T., Jost, A.B., Altner, D., DePaolo, D.J., Eisenhauer, A., Kelley, B.M., Lehrmann, D.J., Paytan, A., Yu, M., Silva-Tamayo, J.C., Payne, J.L., 2017. The influence of seawater carbonate chemistry, mineralogy, and diagenesis on calcium isotope variations in Lower-Middle Triassic carbonate rocks. *Chem. Geol.* 471, 13–37. <https://doi.org/10.1016/j.chemgeo.2017.09.006>.
- Lazar, B., Erez, J., 1990. Extreme ^{13}C depletions in seawater-derived brines and their implications for the past geochemical carbon cycle. *Geology* 18, 1191–1194. [https://doi.org/10.1130/0091-7613\(1990\)018<1191:ECDISD>2.3.CO;2](https://doi.org/10.1130/0091-7613(1990)018<1191:ECDISD>2.3.CO;2).
- Lenton, T.M., Daines, S.J., Mills, B.J.W., 2018. COPSE reloaded: an improved model of biogeochemical cycling over Phanerozoic time. *Earth-Sci. Rev.* 178, 1–28. <https://doi.org/10.1016/j.earscirev.2017.12.004>.
- Melini, L.A., Westphal, H., Swart, P.K., Eberli, G.P., Munnecke, A., 2002. Questioning carbonate diagenetic paradigms: evidence from the Neogene of the Bahamas. *Mar. Geol.* 185, 27–53. [https://doi.org/10.1016/S0025-3227\(01\)00289-4](https://doi.org/10.1016/S0025-3227(01)00289-4).
- Morris, J.L., Puttick, M.N., Clark, J.W., Edwards, D., Kenrick, P., Pressell, S., Wellman, C. H., Yang, Z., Schneider, H., Donoghue, P.C.J., 2018. The timescale of early land plant evolution. *Proc. Natl. Acad. Sci.* 115, E2274–E2283. <https://doi.org/10.1073/pnas.1719588115>.
- Panchuk, K.M., Holmden, C.E., Leslie, S.A., 2006. Local controls on carbon cycling in the Ordovician midcontinent region of north America, with implications for carbon isotope secular curves. *J. Sediment. Res.* 76, 200–211. <https://doi.org/10.2110/jsr.2006.017>.
- Patterson, W.P., Walter, L.M., 1994. Depletion of ^{13}C in seawater ΣCO_2 on modern carbonate platforms: Significance for the carbon isotopic record of carbonates. *Geology* 22, 885–888. [https://doi.org/10.1130/0091-7613\(1994\)022<0885:DOCISC>2.3.CO;2](https://doi.org/10.1130/0091-7613(1994)022<0885:DOCISC>2.3.CO;2).
- Planavsky, N.J., Fakrae, M., Bolton, E.W., Reinhard, C.T., Isson, T.T., Zhang, S., Mills, B.J.W., 2022. On carbon burial and net primary production through Earth's history. *Am. J. Sci.* 322, 413–460. <https://doi.org/10.2475/03.2022.01>.
- Rasmussen, C.M.Ø., Ullmann, C.V., Jakobsen, K.G., Lindskog, A., Hansen, J., Hansen, T., Eriksson, M.E., Dronov, A., Frei, R., Korte, C., Nielsen, A.T., Harper, D.A.T., 2016. Onset of main Phanerozoic marine radiation sparked by emerging Mid Ordovician icehouse. *Sci. Rep.* 6, 18884. <https://doi.org/10.1038/srep18884>.
- Romanek, C.S., Grossman, E.L., Morse, J.W., 1992. Carbon isotopic fractionation in synthetic aragonite and calcite: Effects of temperature and precipitation rate. *Geochim. Cosmochim. Acta* 56, 419–430. [https://doi.org/10.1016/0016-7037\(92\)90142-6](https://doi.org/10.1016/0016-7037(92)90142-6).
- Ross Jr., R.J., James, N.P., Hintze, L.F., Poole, F.G., 1989. Architecture and evolution of a Whiterockian (Early Middle Ordovician) carbonate platform, basin ranges of western U.S.A. Crevello, P.D., Wilson, J.L., Sarg, J.F., Read, J.F. (Eds.). *Controls on Carbonate Platforms and Basin Development*. SEPM Society for Sedimentary Geology. <https://doi.org/10.2110/pec.89.44.0167>, p. 0.
- Ruttenberg, K.C., 2003. 8.13 - the global phosphorus cycle. Holland, H.D., Turekian, K.K. (Eds.). *Treatise on Geochemistry*. Pergamon, Oxford, pp. 585–643. <https://doi.org/10.1016/B0-08-043751-6/08153-6>.
- Saltzman, M.R., Edwards, C.T., 2017. Gradients in the carbon isotopic composition of Ordovician shallow water carbonates: a potential pitfall in estimates of ancient CO_2 and O_2 . *Earth Planet. Sci. Lett.* 464, 46–54. <https://doi.org/10.1016/j.epsl.2017.02.011>.
- Saltzman, M.R., Edwards, C.T., Leslie, S.A., Dwyer, G.S., Bauer, J.A., Repetski, J.E., Harris, A.G., Bergström, S.M., 2014. Calibration of a conodont apatite-based Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ curve to biostratigraphy and geochronology: Implications for stratigraphic resolution. *Geol. Soc. Am. Bull.* 126, 1551–1568. <https://doi.org/10.1130/B31038.1>.
- Saltzman, M.R., Thomas, E., 2012. Carbon isotope stratigraphy. Gradstein, F.M., Ogg, J. G., Schmitz, M., Ogg, G.M. (Eds.). *The Geologic Time Scale*. Elsevier, pp. 207–232.
- Scotese, C.R., Wright, N.M., 2018. PALEOMAP Paleodigital elevation MODELS (PaleoDEMS) for the Phanerozoic. 10.5281/zenodo.5460860.
- Stigall, A.L., Edwards, C.T., Freeman, R.L., Rasmussen, C.M.Ø., 2019. Coordinated biotic and abiotic change during the Great Ordovician Biodiversification Event: Darriwilian assembly of early Paleozoic building blocks. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 530, 249–270. <https://doi.org/10.1016/j.palaeo.2019.05.034>.
- Stiller, M., Rounick, J.S., Shasha, S., 1985. Extreme carbon-isotope enrichments in evaporating brines. *Nature* 316, 434–435. <https://doi.org/10.1038/316434a0>.
- Swanson-Hysell, N.L., Macdonald, F.A., 2017. Tropical weathering of the Taconic orogeny as a driver for Ordovician cooling. *Geology* 45, 719–722. <https://doi.org/10.1130/G38985.1>.
- Swart, P.K., 2008. Global synchronous changes in the carbon isotopic composition of carbonate sediments unrelated to changes in the global carbon cycle. *Proc. Natl. Acad. Sci.* 105, 13741–13745. <https://doi.org/10.1073/pnas.0802841105>.
- Tang, J., Dietzel, M., Böhm, F., Köhler, S.J., Eisenhauer, A., 2008a. $\text{Sr}^{2+}/\text{Ca}^{2+}$ and $^{44}\text{Ca}/^{40}\text{Ca}$ fractionation during inorganic calcite formation: II. Ca isotopes. *Geochim. Cosmochim. Acta* 72, 3733–3745. <https://doi.org/10.1016/j.gca.2008.05.033>.
- Tang, J., Köhler, S.J., Dietzel, M., 2008b. $\text{Sr}^{2+}/\text{Ca}^{2+}$ and $^{44}\text{Ca}/^{40}\text{Ca}$ fractionation during inorganic calcite formation: I. Sr incorporation. *Geochim. Cosmochim. Acta* 72, 3718–3732. <https://doi.org/10.1016/j.gca.2008.05.031>.
- Wang, J., Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., 2023. Application of the $\delta^{44}/^{40}\text{Ca}$ - $^{88}/^{86}\text{Sr}$ multi-proxy to Namibian Marinoan cap carbonates. *Geochim. Cosmochim. Acta* 353, 13–27. <https://doi.org/10.1016/j.gca.2023.04.023>.
- Wang, J., Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., 2021. Stable Ca and Sr isotopes support volcanically triggered biocalcification crisis during Oceanic Anoxic Event 1a. *Geology* 49, 515–519. <https://doi.org/10.1130/G47945.1>.
- Woglsland, B.V., 2020. Organomineralization of microbialites from Storr's Lake. San Salvador Island, Bahamas: calcium stable isotope analysis using TIMS and a ^{42}Ca - ^{43}Ca double spike. The Ohio State University.
- Wu, R., Calner, M., Lehnert, O., 2017. Integrated conodont biostratigraphy and carbon isotope chemostratigraphy in the Lower–Middle Ordovician of southern Sweden reveals a complete record of the MDICE. *Geol. Mag.* 154, 334–353. <https://doi.org/10.1017/S0016756816000017>.